

Development of Advanced Lithographic Processes for Emerging Device Technologies

Timothy Carroll^{1,2}, Kevin Owen¹, Bennet Clark¹, Neal Lewis³ and Matthew Moneck¹

¹Department of Electrical and Computer Engineering, Carnegie Mellon University, Pittsburgh, PA, USA

²College of Marin, Kentfield, CA, USA

³Department of Material Science Engineering, Carnegie Mellon University, Pittsburgh, PA, USA



INTRODUCTION

Objective:

- To explore the potential for developing smaller features than the current minimum feature size of ~1 micron for the Heidelberg MLA, as specified by equipment manufacturers.

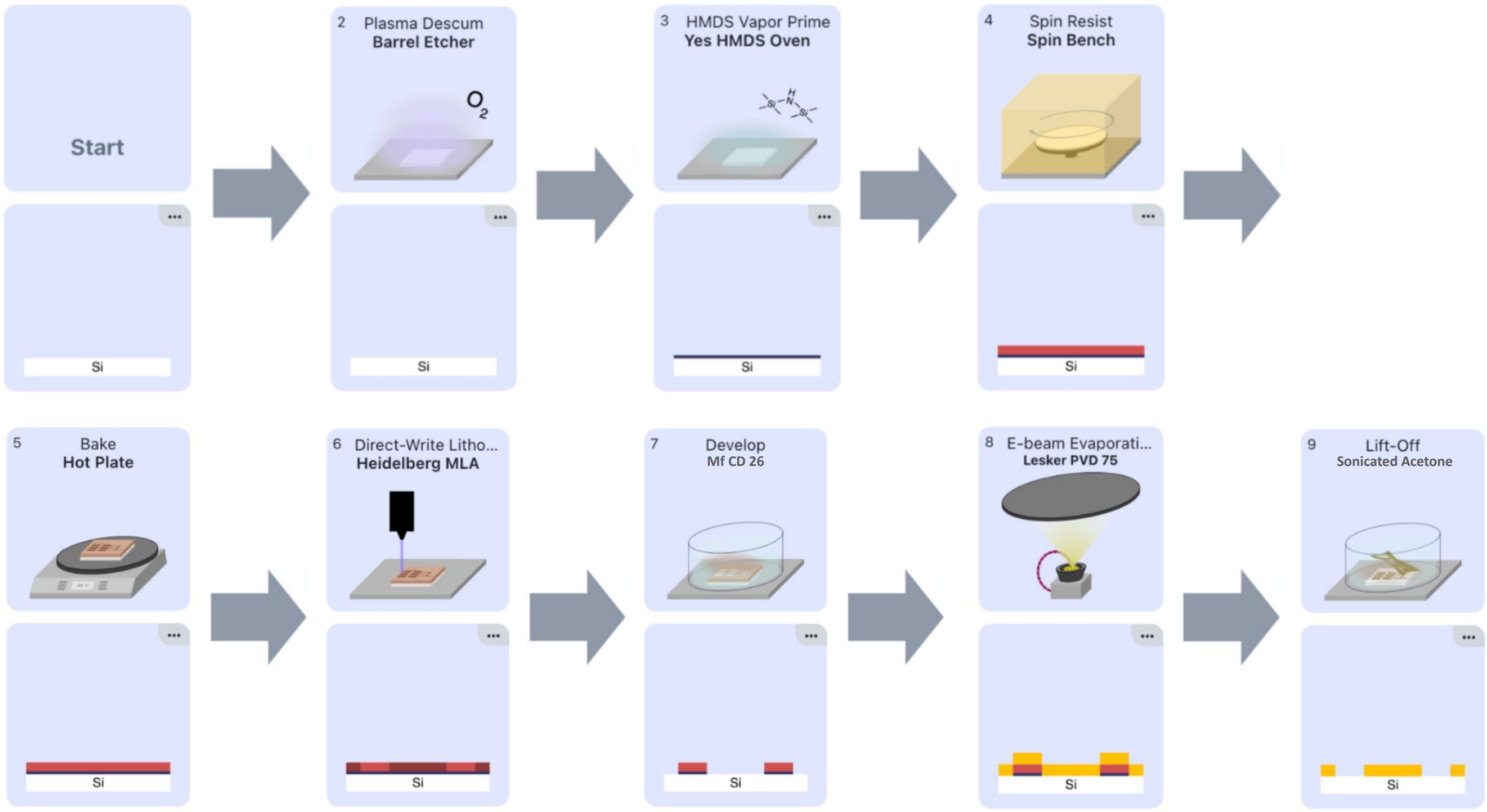
Background:

- Decreasing feature size increases performance density, enabling more computing power in a smaller space.

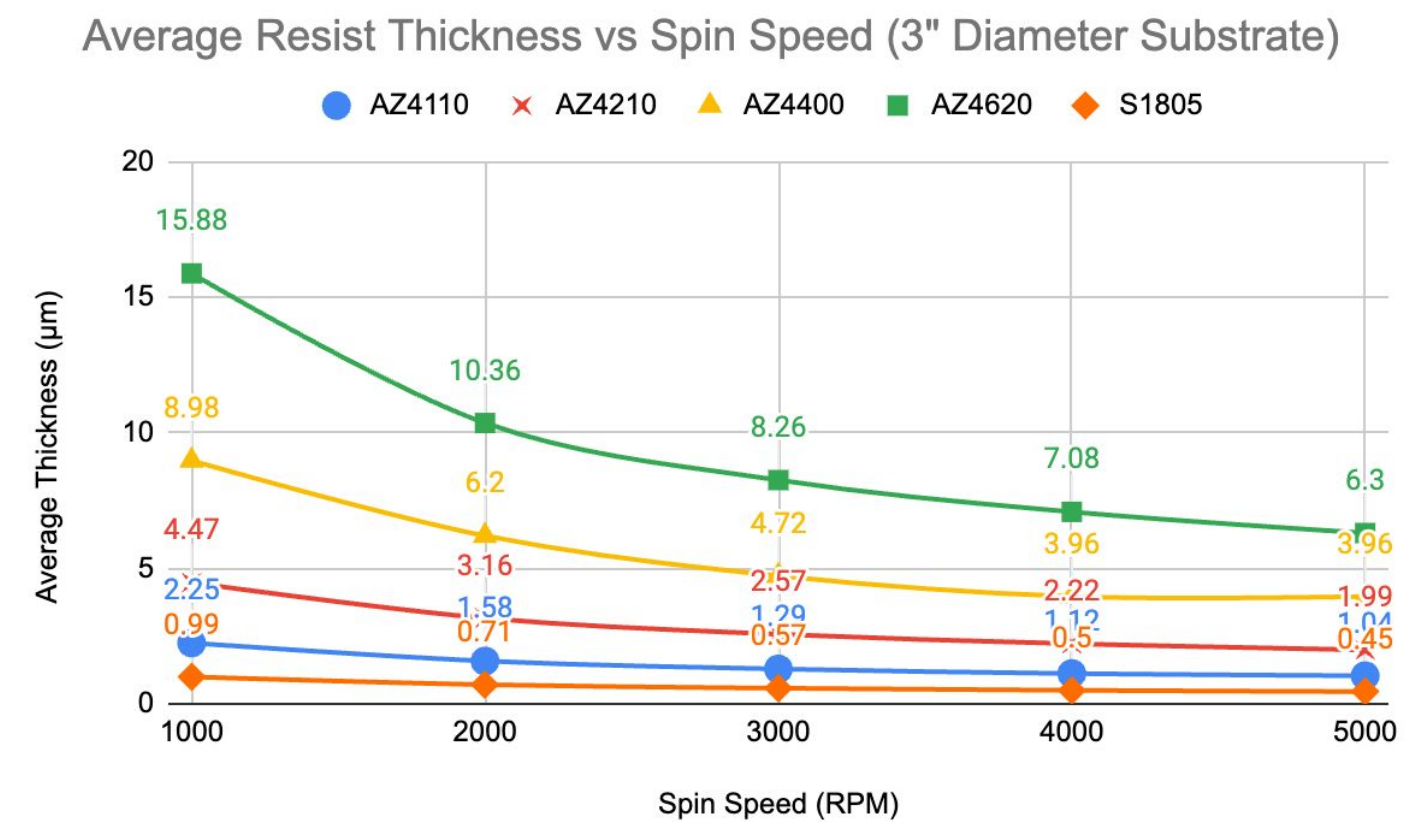
Motivation:

- Enhance the precision and uniformity of nanofabrication processes.
- Advance the miniaturization and performance of electronic devices through innovative techniques.

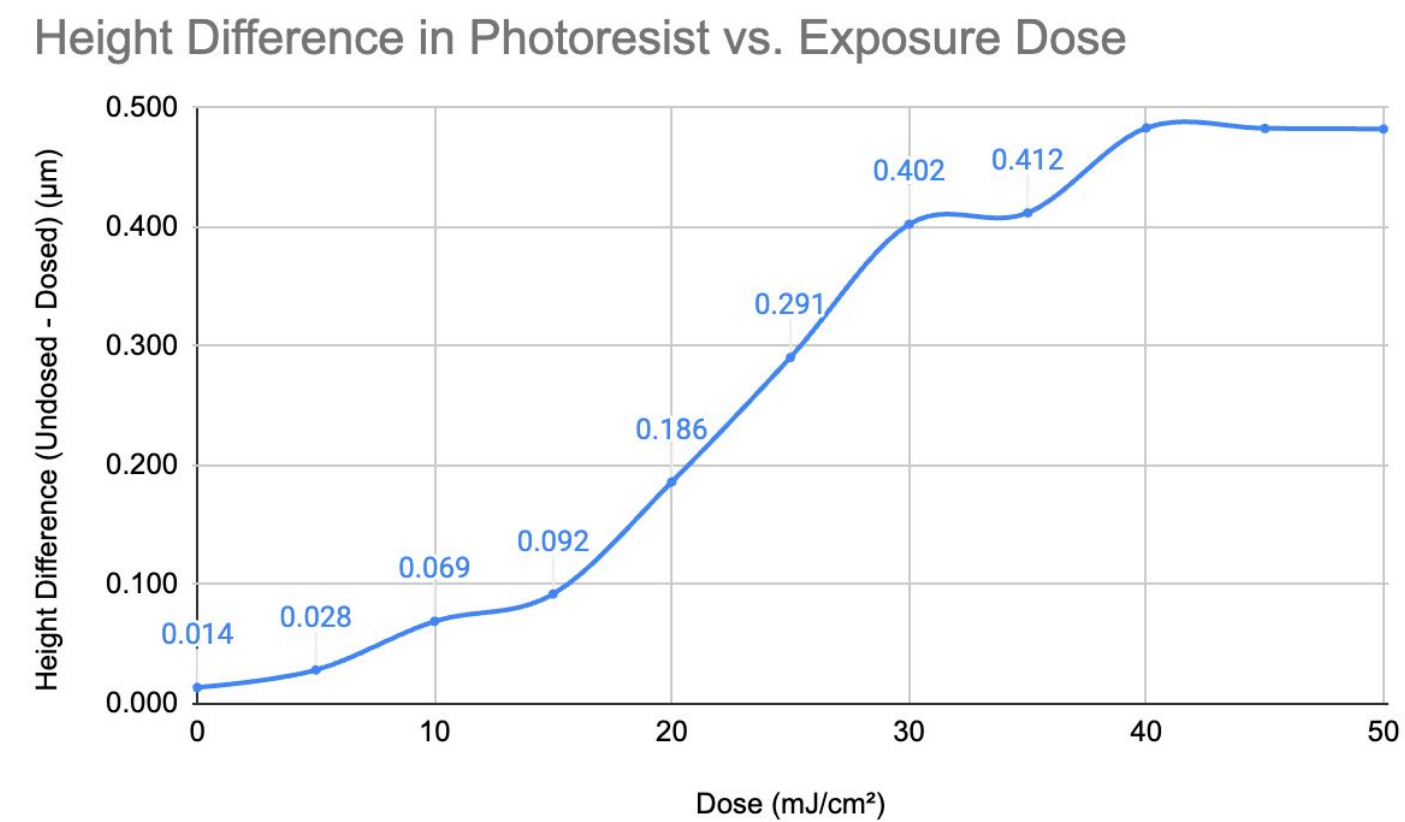
Approach



Preliminary Work



- Lithographic processes fidelity and resolution become increasingly reliant on thickness of resist as feature size decreases due to factors of absorption and reflection.



- Ideally a stepwise function to have either completely developed or completely undeveloped features

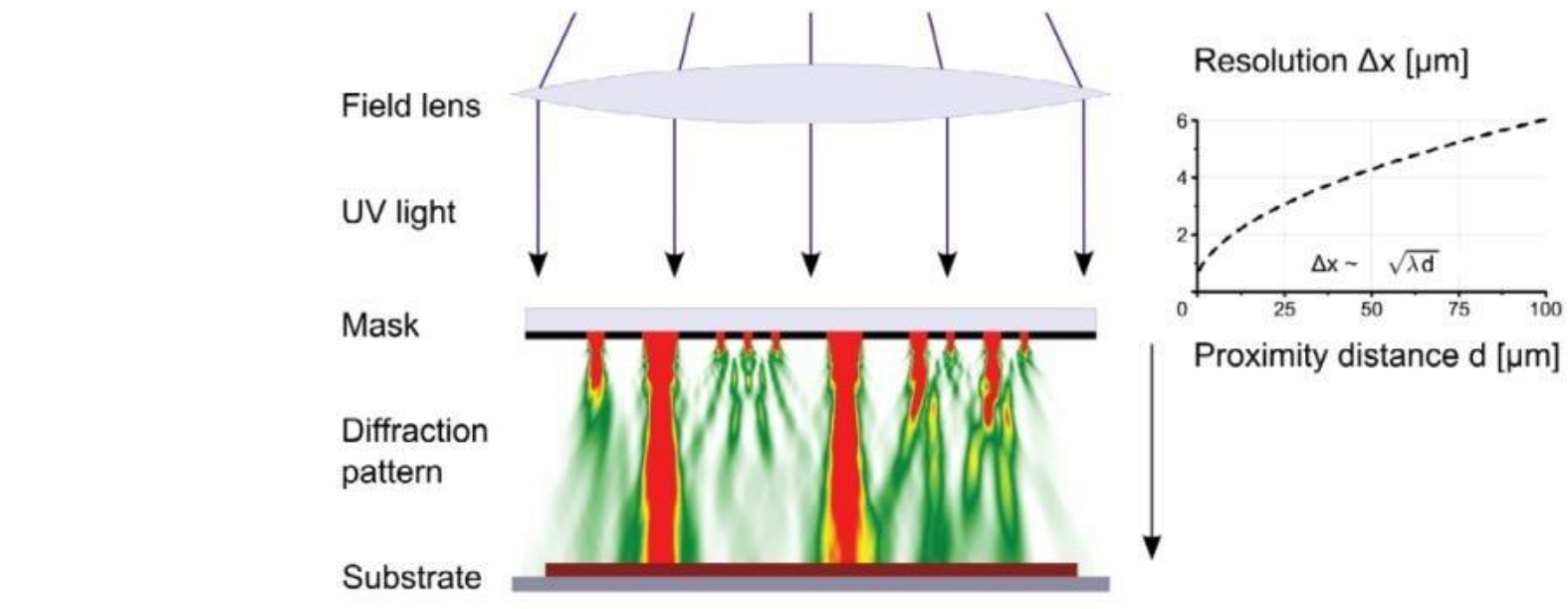
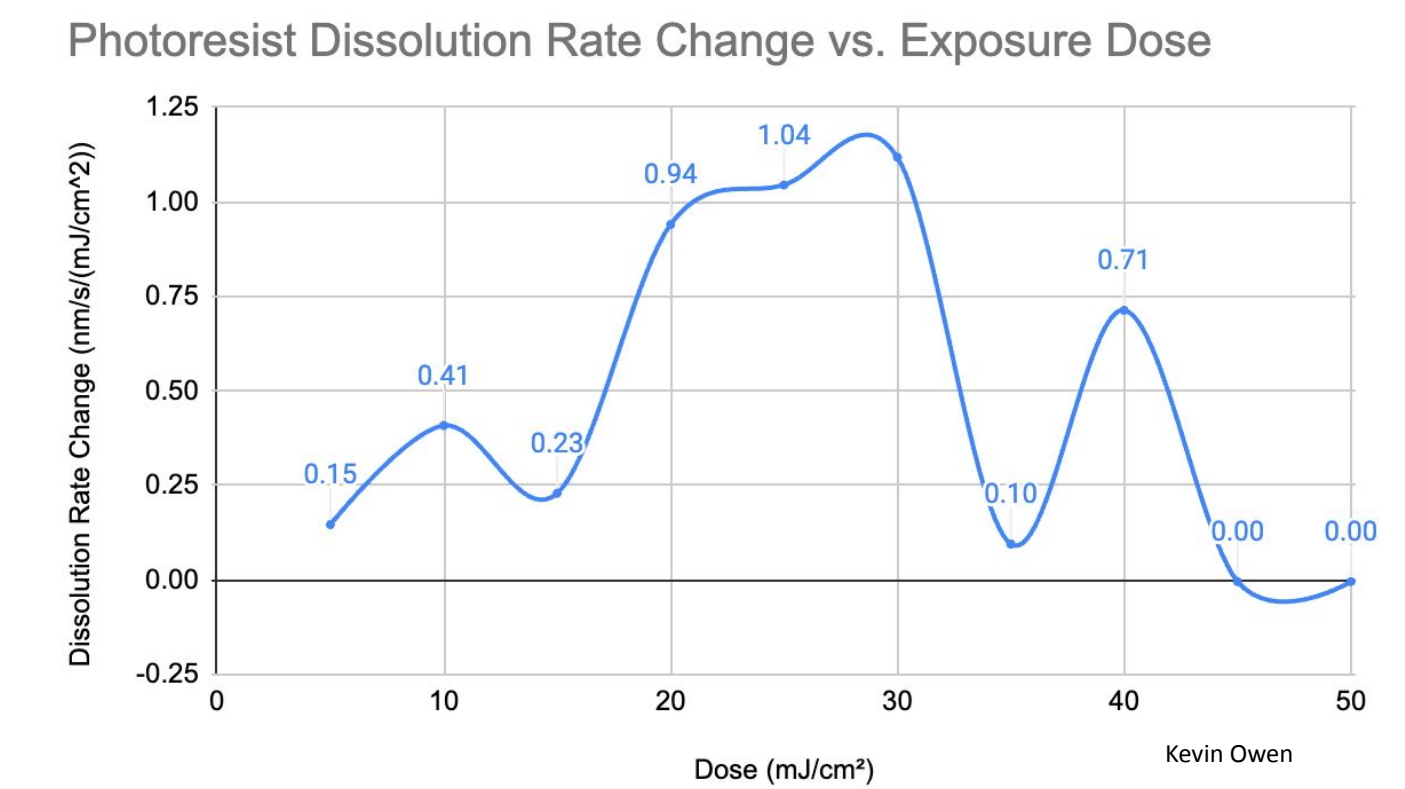
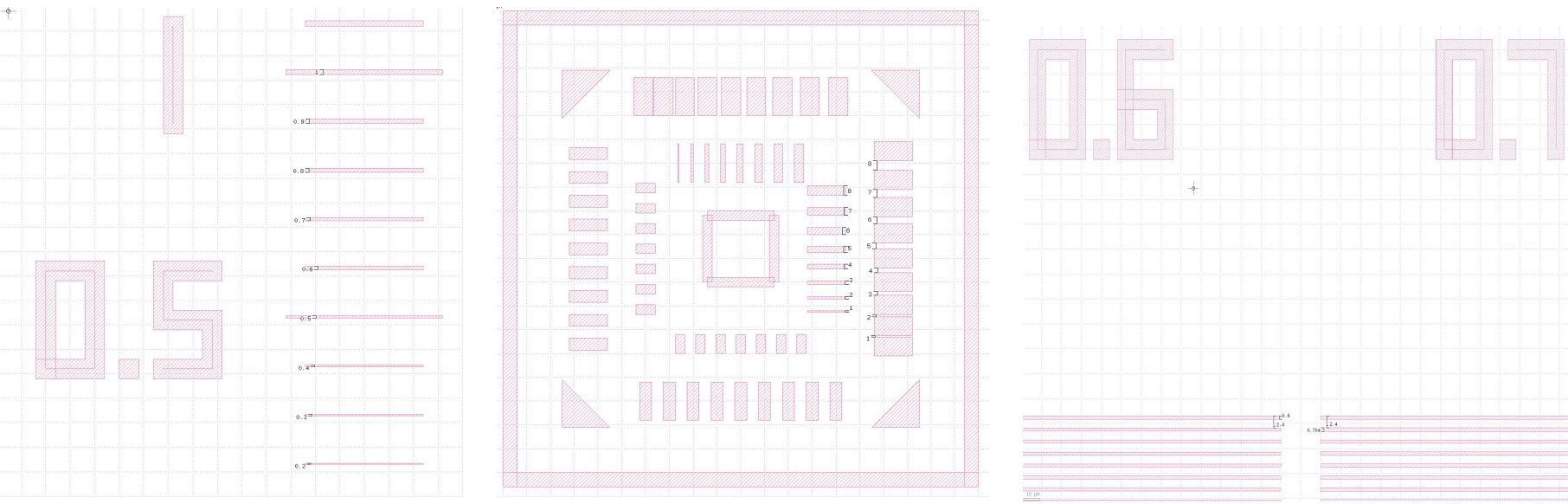
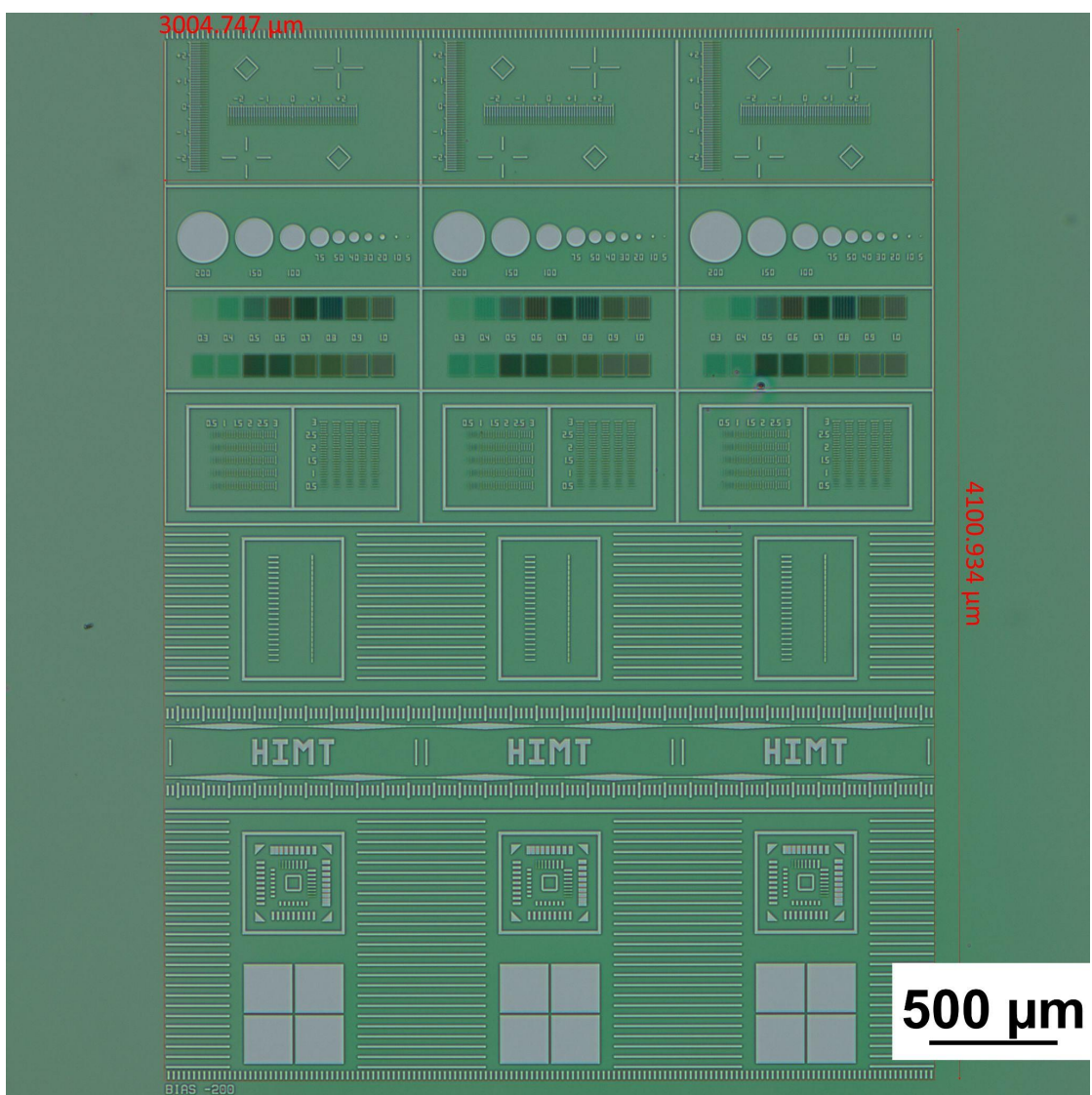


Fig. 1. Trade-off between resolution and yield in conventional shadow printing based mask aligner lithography.
L. Stuerzbecher, F. Fuchs, U. D. Zeitner, and A. Tünnermann, "High-resolution proximity lithography for nano-optical components," *Microelectron. Eng.* 132, 120–134 (2015).

- Light Scattering reducing fidelity proportional to thickness

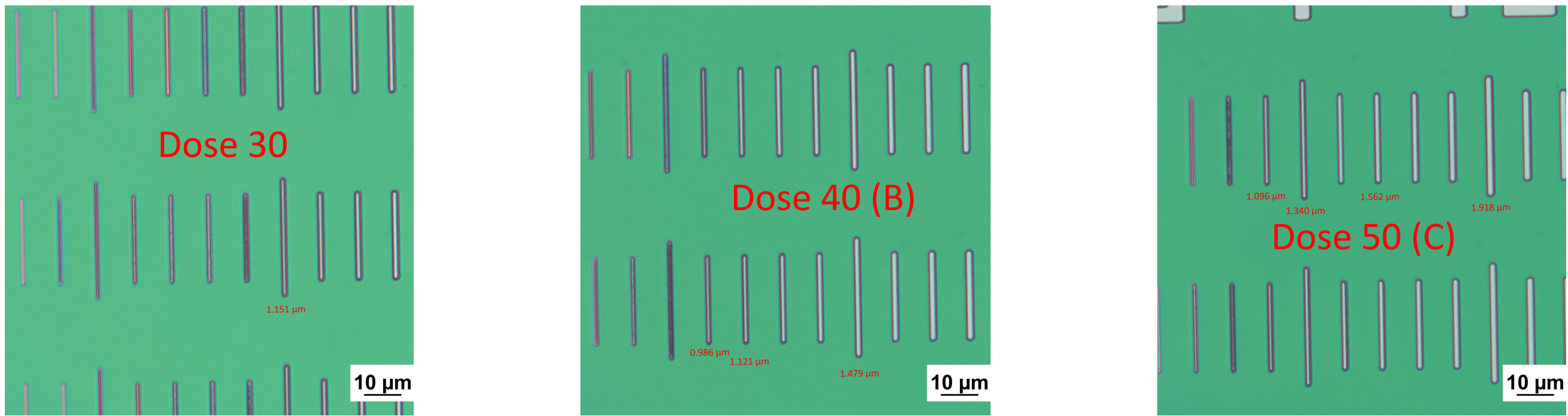


- Features of the mask utilize both size and density to determine efficacy



- Wafer die shot 3mm x 4mm

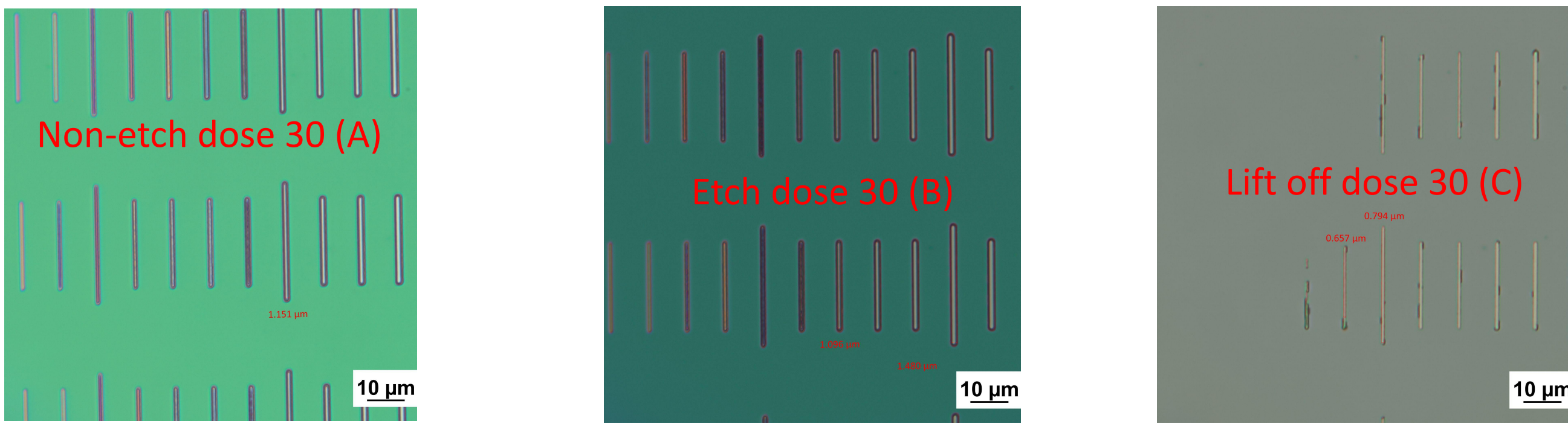
Dose Series



Mask	Dose 30 mJ/cm ²	Dose 40 mJ/cm ²	Dose 50 mJ/cm ²
1.5μm	1.51μm	1.479μm	1.918μm
1.2μm	N/A	1.121μm	1.562μm
1.0μm	N/A	N/A	1.340μm

Under Exposure Matrix

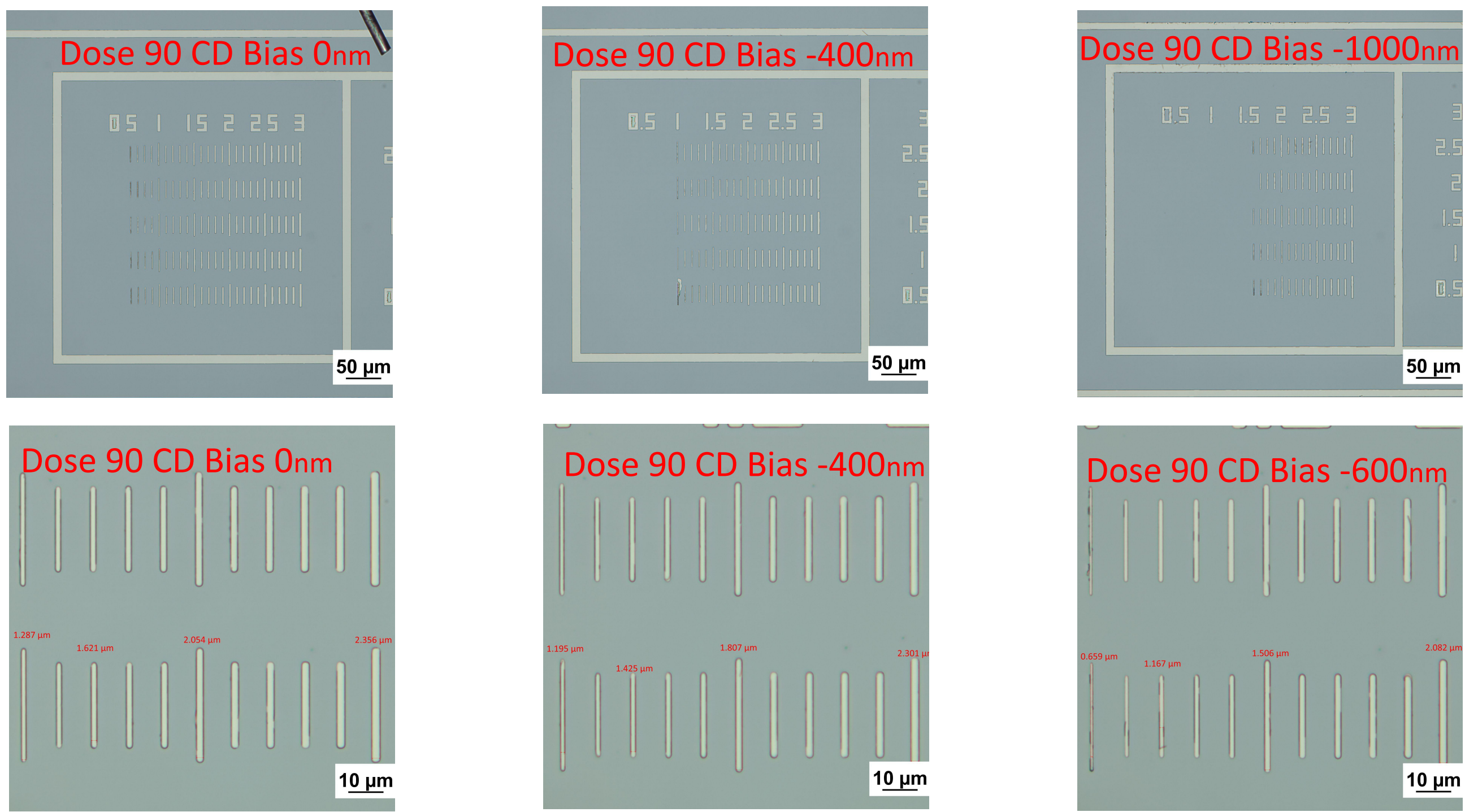
- Underexposing the wafer takes advantage of the exposure process exposing the center of features more than the perimeter
- The underexposure method required a deviation from the standard process by utilizing the barrel etcher for a post-etch



Mask	A	B	C
1.5μm	1.51μm	1.48 μm	0.794μm
1.2μm	N/A	1.096μm	0.657μm
1.0μm	N/A	N/A	N/A

Critical Dimension Bias Reduction Matrix

- CD bias reduction compensates for image widening caused by effects like standing waves and constructive/destructive interference.



Mask	0nm	-400nm	-600nm
1.5μm	2.054	1.807μm	1.506μm
1.2μm	1.621	1.425μm	1.167μm
1.0μm	1.287	1.195μm	0.659μm

Conclusion and Future Work

- Both underexposure with plasma etching and CD bias reduction using the Heidelberg MLA struggle with achieving reliable, repeatable results for high-yield. However these methods hold potential for rapid prototyping and R&D in emerging computing devices where high volume manufacturing is not the priority
- Future work includes imaging with an SEM to characterize sidewalls, attempting the process with different a photoresist/substrate/evaporated material, and optimization of post etch

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